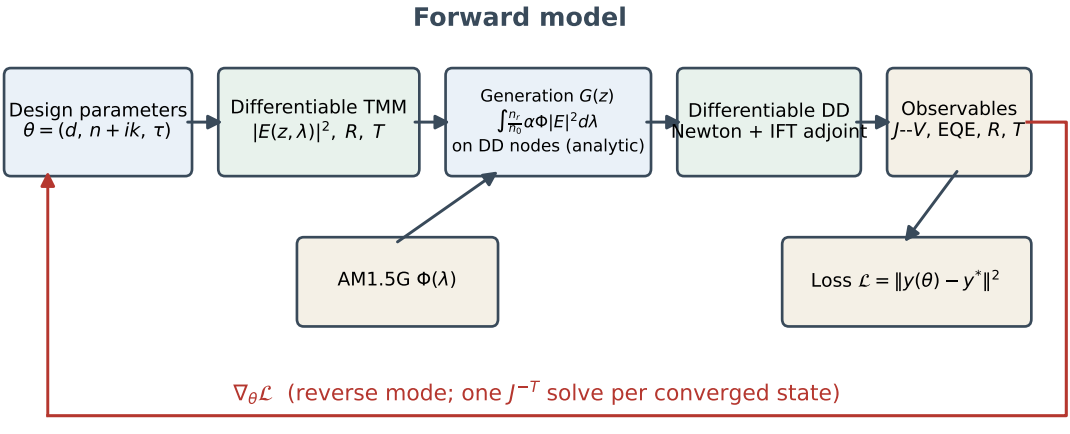
Forward model
Design parameters
$\theta = (d,\ n + ik,\ \tau)$
Differentiable TMM
$|E(z,\lambda)|^2,\ R,\ T$
Generation $G(z)$
$\int_{n_0}^{n_r} \alpha\Phi|E|^2 d\lambda$
on DD nodes (analytic)
Differentiable DD
Newton + IFT adjoint
Observables
J--V, EQE, R, T
AM1.5G $\Phi(\lambda)$
Loss $\mathcal{L} = \|y(\theta) - y^*\|^2$
$\nabla_\theta \mathcal{L}$ (reverse mode; one J^{-T} solve per converged state)